VOLTAGE, CURRENT AND RESISTANCE

***Voltage* / Potential Difference (V):** provides the energy to drive electrons around a circuit.

Measured in - volts (V)

***Current* (I):** a measure of the rate at which charge (electrons) flows in a circuit.

Measured in - Amperes (A)

***Resistance* (R):** a measure of the difficulty electrons have in moving around a circuit.

Measured in - Ohms (Ω)

Voltage equals current times resistance

$$V = IR$$

The ***resistance*** of a wire is dependent on the following.

- ***Type of material used.***

  e.g. Copper has a lower resistance than nichrome wire.

- ***Diameter of the wire.***

  e.g. Thin wire has a greater resistance than thick wire.

The filaments in globes are very thin to give them high resistance which makes the metal get very hot and glow.

- ***Length of the wire.***

  e.g. Long wire has more resistance.

Rheostats (variable resistors) used to control fans and lights have the wire wound around a non-conductor to save space.